

C912

College Algebra

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A filling machine at a bottling plant that runs two shifts targets 500 mL, and a bottle passes if $|x - 500| \leq 4$, where x is volume in mL. The plant ships to 9 states. Which volumes pass?

- A. $496 \leq x \leq 504$
- B. $x \leq 504$
- C. $x \leq 496$ or $x \geq 504$
- D. $492 \leq x \leq 508$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Which inequality describes all numbers x that are at least 3 units away from 7?

- A. $|x - 7| \leq 3$
- B. $|x + 7| \geq 3$
- C. $|x - 3| \geq 7$
- D. $|x - 7| \geq 3$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A store's pricing app, built by a two-person startup, uses $f(x) = 2x + 1$ and $g(x) = x^2$. What is $(f \circ g)(3)$?

- A. 63
- B. 16
- C. 19
- D. 49

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A shipping company founded in 1998 charges $C(x) = 3x + 6$ dollars for a package of x pounds. What is $C^{-1}(x)$?

- A. $x/3 - 6$
- B. $(x + 6)/3$
- C. $1/(3x + 6)$
- D. $(x - 6)/3$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

For $f(x) = 2x + b$, you are told $f^{-1}(10) = 4$. What is b ?

- A. 10
- B. 2
- C. 6
- D. -16

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A revenue model written by an analyst who joined the firm in 2021 is $P(x) = -2(x - 1)^2(x + 3)$. Which describes its graph?

- A. Rises to the left, falls to the right; touches the axis at $x = 1$ and crosses at $x = -3$.
- B. Rises to the left, falls to the right; crosses at $x = 1$ and touches at $x = -3$.
- C. Both ends fall; touches at $x = 1$, crosses at $x = -3$.
- D. Falls to the left, rises to the right; touches at $x = 1$, crosses at $x = -3$.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

Using a free online algebra tool, a teacher divides $P(x) = x^3 - 2x^2 + 4x - 5$ by $(x - 2)$. What is the remainder?

- A. -5
- B. 3
- C. 0, since dividing by a linear factor never leaves a remainder
- D. -29

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

When $P(x) = x^2 + kx + 6$ is divided by $(x + 1)$, the remainder is 2. What is k ?

- A. 7
- B. 2
- C. -5
- D. 5

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A sound engineer at a studio that has recorded 40 albums rewrites $\log(100x^3)$, using base-10 logs. Which expression is equivalent?

- A. $2 + 3 \log x$
- B. $3 \log(100x)$
- C. $6 \log x$
- D. $2 + \log x + 3$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

Given $\log 2 \approx 0.301$ and $\log 3 \approx 0.477$, what is $\log 12$?

- A. About 0.287
- B. About 0.778
- C. About 1.079
- D. It cannot be found without a calculator.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

In a graphing app rated 4.8 stars, a teacher graphs $f(x) = x^2$ and then $g(x) = (x - 4)^2 + 2$. How is g obtained from f ?

- A. Shift right 2 and up 4
- B. Shift right 4 and up 2
- C. Stretch vertically by 4 and shift up 2
- D. Shift left 4 and up 2

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

Solving $\sqrt{x + 6} = x$, a student in an evening class taught by a retired engineer squares both sides and gets $x = 3$ or $x = -2$. What is the solution set?

- A. No solution, since squaring always creates false roots
- B. $\{-2\}$
- C. $\{3\}$
- D. $\{3, -2\}$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A filling machine at a bottling plant ~~that runs two shifts~~ targets 500 mL, and a bottle passes if $|x - 500| \leq 4$, where x is volume in mL. ~~The plant ships to 9 states.~~ Which volumes pass?

Target word: solution set **Decides:** $\leq$ means between $500 - 4$ and $500 + 4$.

A	Correct	Within 4 mL of 500.
B	Dropped element	Drops the lower bound.
C	Inverted operation	That is the $\geq$ (outside) case.
D	Wrong stem number	Uses ± 8 instead of ± 4 .

2. Answer D Concept 1 • transfer

Which inequality describes all numbers x that are at least 3 units away from 7?

Target word: write the inequality **Decides:** “At least 3 away” is distance ≥ 3 .

A	Inverted operation	That is within 3, not at least 3 away.
B	Confusion set	Centers the distance at -7 .
C	Wrong stem number	Swaps the center and the distance.
D	Correct	Distance from 7 is at least 3.

3. Answer C Concept 2

A store’s pricing app, ~~built by a two-person startup,~~ uses $f(x) = 2x + 1$ and $g(x) = x^2$. What is $(f \circ g)(3)$?

Target word: $f(g(3))$ **Decides:** g first: $g(3) = 9$.

A	Confusion set	$f(3) \times g(3)$ multiplies instead of composing.
B	Adjacent category	$f(3) + g(3)$ adds the functions.
C	Correct	$f(9) = 2(9) + 1$.
D	Inverted operation	$g(f(3)) = 7^2$ applies f first.

4. Answer D Concept 3

A shipping company ~~founded in 1998~~ charges $C(x) = 3x + 6$ dollars for a package of x pounds. What is $C^{-1}(x)$?

Target word: inverse function **Decides:** Undo “times 3, plus 6” in reverse order.

A	Dropped element	Undoes the steps in the wrong order, so the 6 is never divided by 3.
B	Inverted operation	Adds 6 instead of subtracting it.
C	Confusion set	The reciprocal is not the inverse.
D	Correct	Subtract 6, then divide by 3.

5. Answer B Concept 3 • transfer

For $f(x) = 2x + b$, you are told $f^{-1}(10) = 4$. What is b ?

Target word: solve for a parameter **Decides:** $f^{-1}(10) = 4$ means $f(4) = 10$.

A	Wrong stem number	Reports the output, not b .
B	Correct	$2(4) + b = 10$.
C	Dropped element	$10 - 4$ drops the coefficient 2.
D	Inverted operation	Uses $f(10) = 4$, the reversed pair.

6. Answer A Concept 4

A revenue model ~~written by an analyst who joined the firm in 2021~~ is $P(x) = -2(x - 1)^2(x + 3)$. Which describes its graph?

Target word: end behavior and zero behavior **Decides:** Leading term $-2x^3$; zero at 1 has multiplicity 2, at -3 multiplicity 1.

A	Correct	Odd degree, negative lead; even multiplicity touches.
B	Confusion set	Swaps touch and cross.
C	Dropped element	Drops the odd degree, which makes the ends go opposite ways.
D	Inverted operation	Ignores the negative leading coefficient.

7. Answer B Concept 5

~~Using a free online algebra tool,~~ a teacher divides $P(x) = x^3 - 2x^2 + 4x - 5$ by $(x - 2)$. What is the remainder?

Target word: remainder **Decides:** Remainder = $P(2)$.

A	Wrong stem number	The constant term is $P(0)$, not $P(2)$.
B	Correct	$8 - 8 + 8 - 5$.
C	Absolute language	Only a true factor leaves remainder 0.
D	Inverted operation	$P(-2)$ uses the wrong sign.

8. Answer D Concept 5 • transfer

When $P(x) = x^2 + kx + 6$ is divided by $(x + 1)$, the remainder is 2. What is k ?

Target word: solve for k **Decides:** $P(-1) = 2$.

A	Dropped element	Sets the remainder to 0, dropping the given 2.
B	Wrong stem number	Reports the remainder.
C	Inverted operation	Uses $P(1)$ instead of $P(-1)$.
D	Correct	$1 - k + 6 = 2$.

9. Answer A Concept 6

A sound engineer ~~at a studio that has recorded 40 albums~~ rewrites $\log(100x^3)$, using base-10 logs. Which expression is equivalent?

Target word: expanded form **Decides:** Product rule, then power rule.

A	Correct	$\log 100 + \log x^3 = 2 + 3 \log x$.
B	Confusion set	Applies the exponent to 100 as well.
C	Inverted operation	Multiplies the pieces instead of adding them.
D	Adjacent category	Treats the power rule as adding the exponent.

10. Answer C Concept 6 • transfer

Given $\log 2 \approx 0.301$ and $\log 3 \approx 0.477$, what is $\log 12$?

Target word: evaluate by building 12 **Decides:** $12 = 2 \times 2 \times 3$.

A	Inverted operation	Multiplies the logs instead of adding.
B	Dropped element	$\log 2 + \log 3 = \log 6$; one factor of 2 is missing.
C	Correct	$2(0.301) + 0.477$.
D	Absolute language	The product rule builds it from the given values.

11. Answer B Concept 7

In a graphing app ~~rated 4.8 stars~~, a teacher graphs $f(x) = x^2$ and then $g(x) = (x - 4)^2 + 2$. How is g obtained from f ?

Target word: transformations **Decides:** $x - 4$ inside; $+2$ outside.

A	Confusion set	Swaps the horizontal and vertical amounts.
B	Correct	Inside shift is opposite the sign; outside shift matches.
C	Adjacent category	A stretch needs a factor multiplying the function.
D	Inverted operation	$x - 4$ shifts right, not left.

12. Answer C Concept 8

Solving $\sqrt{x + 6} = x$, a student ~~in an evening class taught by a retired engineer~~ squares both sides and gets $x = 3$ or $x = -2$. What is the solution set?

Target word: true solutions **Decides:** Check each: $\sqrt{9} = 3$ works; $\sqrt{4} = 2 \neq -2$.

A	Absolute language	Squaring can add false roots; it does not remove true ones.
B	Inverted operation	Keeps the false root and drops the real one.
C	Correct	-2 is extraneous; a principal square root is never negative.
D	Dropped element	Drops the check in the original equation.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owled.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	7	
Wrong stem number	A figure copied from the prompt that is not the answer.	5	
Adjacent category	The right idea in the wrong bucket.	3	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	12	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	6	
Camouflage	Answers a stem fact that was never the question.	0	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Absolute-value inequalities	$ x - a \leq b$ means $a - b \leq x \leq a + b$ ("between"); $ x - a \geq b$ means $x \leq a - b$ or $x \geq a + b$ ("outside").	Reversing between and outside, or solving only one side.	A tolerance, a distance from a target, or "within".
2	Function composition order	$(f \circ g)(x) = f(g(x))$: apply g first, then f .	Applying f first, or multiplying the functions.	An open circle between two function names.
3	Inverse functions	To find f^{-1} , swap x and y and solve; $f^{-1}(a) = b$ means $f(b) = a$. f^{-1} is not $1/f$.	Writing the reciprocal, or undoing steps in the wrong order.	The $^{-1}$ superscript on a function name.
4	Polynomial end behavior and multiplicity	Degree and leading coefficient set end behavior; a zero of even multiplicity touches the x -axis and one of odd multiplicity crosses it.	Ignoring the sign of the leading term or mixing touch and cross.	A factored polynomial with repeated factors.
5	Remainder theorem	The remainder when $P(x)$ is divided by $(x - c)$ equals $P(c)$.	Evaluating at $-c$, or using the constant term.	"Divided by $(x - c)$, the remainder is..."
6	Properties of logarithms	$\log(ab) = \log a + \log b$, $\log(a/b) = \log a - \log b$, $\log(a^n) = n \log a$; there is no rule for $\log(a + b)$.	Multiplying logs, or applying an exponent to the whole product.	Expand, condense, or evaluate using given log values.
7	Graph transformations	$f(x - h)$ shifts right h ; $f(x) + k$ shifts up k ; a factor outside stretches vertically.	Shifting left for $x - h$, or swapping horizontal and vertical.	A parent function rewritten with numbers inside and outside.
8	Extraneous solutions	Squaring both sides or clearing denominators can create false roots; every candidate must be checked in the original equation.	Keeping every root of the transformed equation.	A square root or a variable in a denominator.

Course scope (public description): This course provides further application and analysis of algebraic concepts and functions through mathematical modeling of real-world situations. Topics include real numbers, algebraic expressions, equations and inequalities, graphs and functions, polynomial and rational functions, exponential and logarithmic functions, and systems of linear equations.